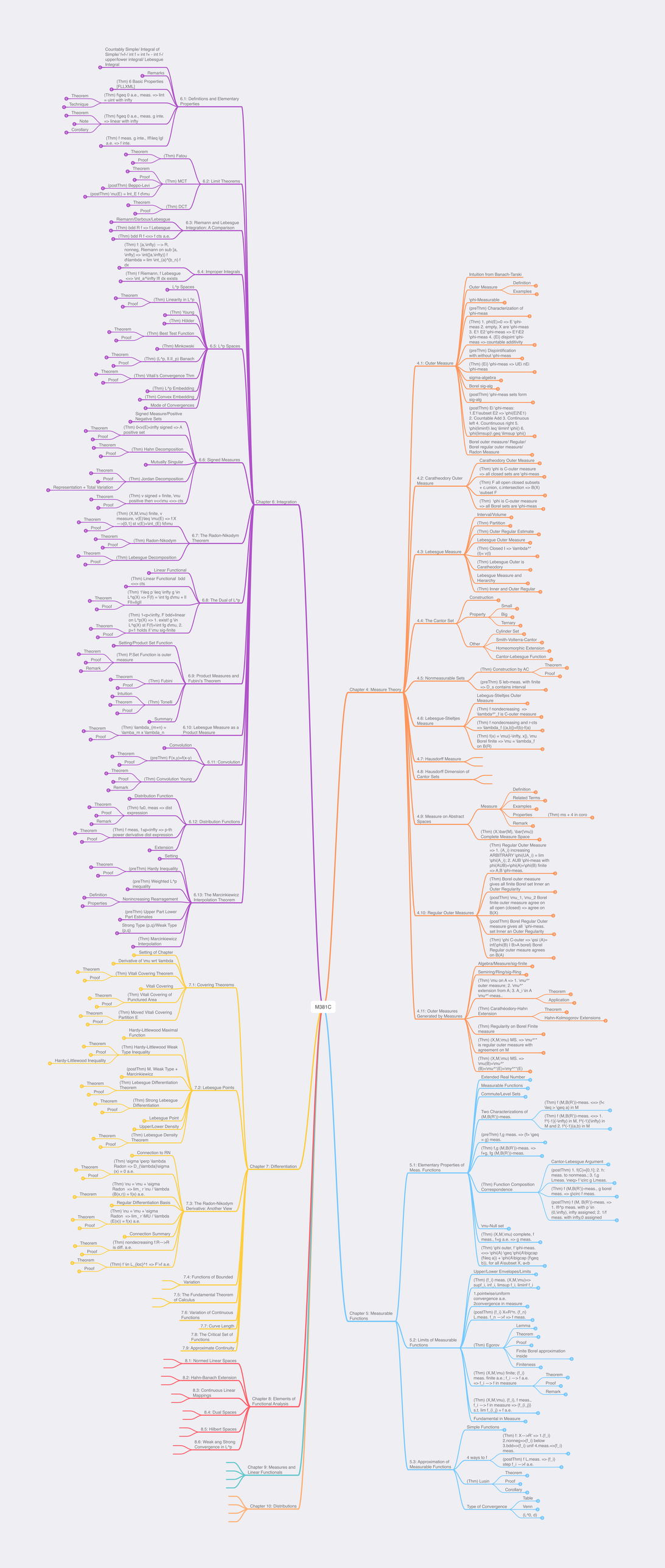
M381C
Chapter 4: Measure Theory
4.1: Outer Measure
Intuition from Banach-Tarski
Outer Measure
Definition
Examples
\phi-Measurable
(preThm) Characterization of \phi-meas
(Thm) 1. phi(E)=0 => E \phi-meas 2. empty, X are \phi-meas 3. E1 E2 \phi-meas => E1\E2 \phi-meas 4. (E) disjoint \phi-meas => countable additivity
(preThm) Disjointification with.without \phi-meas
(Thm) (Ei) \phi-meas => UEi nEi \phi-meas
sigma-algebra
Borel sig-alg
(postThm) \phi-meas sets form sig-alg
(postThm) Ei \phi-meas: 1.E1\subset E2 => \phi(E2\E1) 2. Countable Add 3. Continuous left 4. Countinuous right 5. \phi(liminf\) leq \liminf \phi\) 6. \phi(limsup\) geq \limsup \phi\)
Borel outer measure/ Regular/ Borel regular outer measure/ Radon Measure
4.2: Caratheodory Outer Measure
Caratheodory Outer Measure
(Thm) \phi is C-outer measure => all closed sets are \phi-meas
(Thm) F all open closed subsets + c.union, c.intersection => B(X) \subset F
(Thm) \phi is C-outer measure => all Borel sets are \phi-meas
4.3: Lebesgue Measure
Interval/Volume
(Thm) Partition
(Thm) Outer Regular Estimate
Lebesgue Outer Measure
(Thm) Closed I => \lambda^* (I)= v(I)
(Thm) Lebesgue Outer is Caratheodory
Lebesgue Measure and Hierarchy
(Thm) Inner and Outer Regular
4.4: The Cantor Set
Construction
Property
Small
Big
Ternary
Other
Cylinder Set
Smith-Volterra-Cantor
Homeomorphic Extension
Cantor-Lebesgue Function
4.5: Nonmeasurable Sets
(Thm) Construction by AC
Theorem
Proof
(preThm) S leb-meas. with finite => D_s contains interval
4.6: Lebesgue-Stieltjes Measure
Lebegus-Stieltjes Outer Measure
(Thm) f nondecreasing => \lambda^*_f is C-outer measure
(Thm) f nondecreasing and r-cts => \lambda_f ((a,b])=f(b)-f(a)
(Thm) f(x) = \mu((-\infty, x]), \mu Borel finite => \mu = \lambda_f on B(R)
4.7: Hausdorff Measure
4.8: Hausdorff Dimension of Cantor Sets
4.9: Measure on Abstract Spaces
Measure
Definition
Related Terms
Examples
Properties
(Thm) ms + 4 in coro
Remark
(Thm) (X,\barₘ(M), \barₘ(\mu)) Complete Measure Space
4.10: Regular Outer Measures
(Thm) Regular Outer Measure => 1. (A_i) incressing ARBITRARY \phi(\UA_i) = lim \phi(A_i); 2. AUB \phi-meas with phi(AUB)+\phi(A)+\phi(B) finite => A,B \phi-meas.
(Thm) Borel outer measure gives all finite Borel set Inner an Outer Regularity
(postThm) \mu_1, \mu_2 Borel finite outer measure agree on all open (closed) => agree on B(X)
(postThm) Borel Regular Outer measure gives all \phi-meas. set Inner an Outer Regularity
(Thm) \phi C-outer => \psi (A)= inf{\phi(B) | B⊃A bore} Borel Regular outer meaure agrees on B(A)
4.11: Outer Measures Generated by Measures
Algebra/Measure/sig-finite
Semiring/Ring/sig-Ring
(Thm) \mu on A => 1. \mu^* outer measure; 2. \mu^* extension from A; 3. A_i \in A \mu^*-meas..
Theorem
Application
(Thm) Caratheodory-Hahn Extension
Theorem
Hahn-Kolmogorov Extensions
(Thm) Regularity on Borel Finite measure
(Thm) (X,M,\mu) MS. => \mu^** is regular outer measure with agreement on M
(Thm) (X,M,\mu) MS. => \mu(B)=\mu^*(B)=\mu^**(E)
Chapter 5: Measurable Functions
5.1: Elementary Properties of Meas. Functions
Extended Real Number
Measurable Functions
Commute/Level Sets
Two Characterizations of (M,B(R))-meas.
(Thm) f (M,B(R))-meas. <=> 1. f^(-1)(-\infty) in M, f^(-1)(\infty) in M and 2. f^(-1)(a,b) in M
(preThm) f,g meas. => (f> \geq = g) meas.
(Thm) f,g (M,B(R))-meas. => f+g, fg (M,B(R))-meas.
(Thm) Function Composition Correspondence
Cantor-Lebesgue Argument
(postThm) 1. f(C)=[0,1]; 2. h: meas. to nonmeas.; 3. f,g Lmeas. \neq> f \circ g Lmeas.
(Thm) f (M,B(R))-meas., g borel meas. => g\circ f meas.
(postThm) f (M, B(R))-meas. => 1. lf|^p meas. with p \in (0,\infty), inty assigned; 2. 1/f meas. with inty,0 assigned
\mu-Null set
(Thm) (X,M,\mu) complete, f meas., f=g a.e. => g meas.
(Thm) \phi outer, f \phi-meas. <=> \phi(A) \geq \phi(A\bigcap (f\leq a)) + \phi(A\bigcap (f\geq b)), for all A\subset X, a<b
5.2: Limits of Measurable Functions
Upper/Lower Envelopes/Limits
(Thm) {f_i} meas. (X,M,\mu)=> sup f_i, inf_i, limsup f_i, liminf f_i
1.pointwise/uniform convergence a.e. 2convergence in measure
(postThm) {f_i} X=R^n, (f_n) L-meas, f_n --> f => f meas.
(Thm) Egorov
Lemma
Theorem
Proof
Finite Borel approximation inside
Finiteness
(Thm) (X,M,\mu) finite, {f_i} meas. finite a.e., f_i --> f a.e. => f_i --> f in measure
Theorem
Proof
Remark
(Thm) (X,M,\mu), {f_i}, f meas., f_i --> f in measure => {f_{i_j}} s.t. lim f_{i_j} = f a.e.
Fundamental in Measure
5.3: Approximation of Measurable Functions
Simple Functions
(Thm) f: X--->R' => 1.{f_i} 2.nonneg=>{f_i} below 3.bdd=>f_i} unif 4.meas.=>{f_i} meas.
4 ways to f
(postThm) f L-meas. => {f_i} step f_i --> f a.e.
(Thm) Lusin
Theorem
Proof
Corollary
Type of Convergence
Table
Venn
(L^0, d)
Chapter 6: Integration
6.1: Definitions and Elementary Properties
Countably Simple/ Integral of Simple/ f+f-/ int f = int f+ - int f-/ upper/lower integral/ Lebesgue Integral
Remarks
(Thm) 6 Basic Properties [FLLXML]
(Thm) f\geq 0 a.e., meas. => lint = uint with infty
Theorem
Technique
(Thm) f\geq 0 a.e., meas. g inte. => linear with infty
Theorem
Note
Corollary
(Thm) f meas. g inte., |f|\leq |g| a.e. => f inte.
6.2: Limit Theorems
(Thm) Fatou
Theorem
Proof
(Thm) MCT
Theorem
Proof
(postThm) Beppo-Levi
(postThm) \nu(E) = Int_E f d\mu
(Thm) DCT
Theorem
Proof
6.3: Riemann and Lebesgue Integration: A Comparison
Riemann/Darboux/Lebesgue
(Thm) bdd R f => f Lebesgue
(Thm) bdd R f <=> f cts a.e.
6.4: Improper Integrals
(Thm) f: [a,\infty) --> R, nonneg, Riemann on sub [a, \infty) => \int_{[a,\infty)} f d\lambda = lim \int_{a}^{b_n} f dx
(Thm) f Riemann, f Lebesgue <=> \int_a^\infty |f| dx exists
6.5: L^p Spaces
L^p Spaces
(Thm) Linearity in L^p
Theorem
Proof
(Thm) Young
(Thm) Hölder
(Thm) Best Test Function
Theorem
Proof
(Thm) Minkowski
(Thm) (L^p, ||.||_p) Banach
Theorem
Proof
(Thm) Vitali's Convergence Thm
Theorem
Proof
(Thm) L^p Embedding
(Thm) Convex Embedding
Mode of Convergences
6.6: Signed Measures
Signed Measure/Positive Negative Sets
(Thm) 0<v(E)<infty signed => A positive set
Theorem
Proof
(Thm) Hahn Decomposition
Theorem
Proof
Mutually Singular
(Thm) Jordan Decomposition
Theorem
Proof
Representation + Total Variation
(Thm) v signed + finite, \mu positive then v<<\mu <=> cts
Theorem
Proof
6.7: The Radon-Nikodym Theorem
(Thm) (X,M,\mu) finite, v measure, v(E)\leq \mu(E) => f:X -->[0,1] st. v(E)=\int_{E} fd\mu
Theorem
Proof
(Thm) Radon-Nikodym
Theorem
Proof
(Thm) Lebesgue Decomposition
Theorem
Proof
6.8: The Dual of L^p
Linear Functional
(Thm) Linear Functional bdd <=> cts
Theorem
Proof
(Thm) 1\leq p \leq \infty g \in L^q(X) => F(f) = \int fg d\mu + lI
(Thm) 1<p<\infty, F bdd+linear on L^p(X) => 1. exist! g \in L^q(X) st F(f)=\int fg d\mu, 2. p=1 holds if \mu sig-finite
Theorem
Proof
6.9: Product Measures and Fubini's Theorem
Setting/Product Set Function
(Thm) P.Set Function is outer measure
Theorem
Proof
Remark
(Thm) Fubini
Theorem
Proof
Intuition
(Thm) Tonelli
Theorem
Proof
Summary
6.10: Lebesgue Measure as a Product Measure
(Thm) \lambda_{m+n} = \lambₘ_m x \lambda_n
Theorem
Proof
6.11: Convolution
Convolution
(preThm) F(x,y)=f(x-y)
Theorem
Proof
(Thm) Convolution Young
Theorem
Proof
Remark
6.12: Distribution Functions
Distribution Function
(Thm) fa0, meas => dist expression
Theorem
Proof
Remark
(Thm) f meas, 1\leq p<\infty => p-th power derivative dist expression
Theorem
Proof
Extension
6.13: The Marcinkiewicz Interpolation Theorem
Setting
(preThm) Hardy Inequality
Theorem
Proof
(preThm) Weighted L^p inequality
Definition
Properties
Nonincreasing Rearrageament
(preThm) Upper Part Lower Part Estimates
Strong Type (p,q)/Weak Type (p,q)
(Thm) Marcinkiewicz Interpolation
Chapter 7: Differentiation
7.1: Covering Theorems
Setting of Chapter
Derivative of \mu wrt \lambda
(Thm) Vitali Covering Theorem
Theorem
Proof
Vitali Covering
(Thm) Vitali Covering of Punctured Area
Theorem
Proof
(Thm) Moved Vitali Covering
Partition E
Theorem
Proof
7.2: Lebesgue Points
Hardy-Littlewood Maximal Function
(Thm) Hardy-Littlewood Weak Type Inequality
Theorem
Proof
Hardy-Littlewood Inequality
(postThm) M. Weak Type + Marcinkiewicz
(Thm) Lebesgue Differentiation Theorem
Theorem
Proof
(Thm) Strong Lebesgue Differentiation
Theorem
Proof
Lebesgue Point
Upper/Lower Density
(Thm) Lebesgue Density Theorem
Theorem
Proof
7.3: The Radon-Nikodym Derivative: Another View
Connection to RN
(Thm) \sigma \perp \lambda
Radon => D_{\lambda}\sigma (x) = 0 a.e.
Theorem
Proof
(Thm) \nu = \nu + \sigma Radon => lim_ r \nu / \lambda (B(x,r)) = f(x) a.e.
Theorem
Regular Differentiation Basis
(Thm) \nu = \nu + \sigma Radon => lim_ r \MU / \lambda (E(x)) = f(x) a.e.
Theorem
Proof
Connection Summary
(Thm) nondecreasing f:R-->R is diff. a.e.
Theorem
Proof
(Thm) f \in L_{loc}^1 => F=f a.e.
Theorem
Proof
7.4: Functions of Bounded Variation
7.5: The Fundamental Theorem of Calculus
7.6: Variation of Continuous Functions
7.7: Curve Length
7.8: The Critical Set of Functions
7.9: Approximate Continuity
Chapter 8: Elements of Functional Analysis
8.1: Normed Linear Spaces
8.2: Hahn-Banach Extension
8.3: Continuous Linear Mappings
8.4: Dual Spaces
8.5: Hilbert Spaces
8.6: Weak ang Strong Convergence in L^p
Chapter 9: Measures and Linear Functionals
Chapter 10: Distributions